

```
import pandas as pd
import matplotlib.pyplot as plt

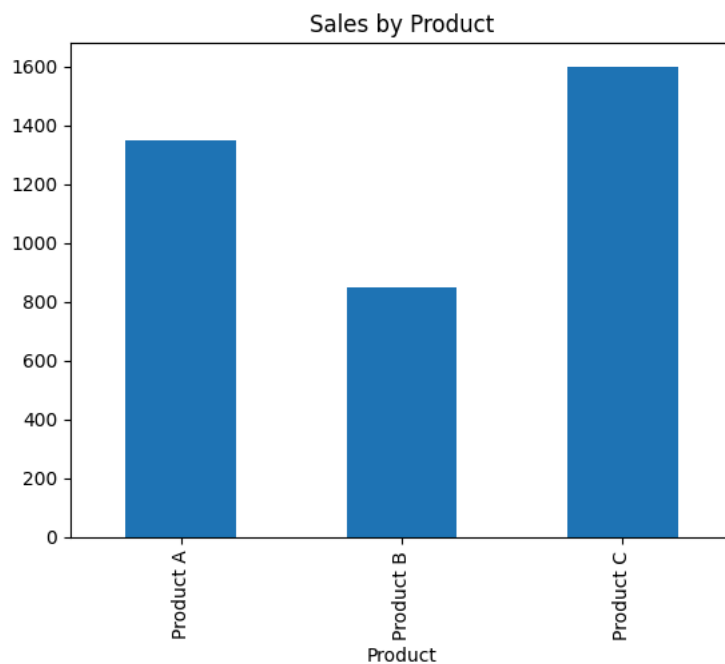
df = pd.read_excel('/content/sales_data.xlsx')
print(df)
```

	Date	Product	Sales	Quantity	Region
0	2023-01-01 00:00:00	Product A	200	4	North
1	2023-02-01 00:00:00	Product B	150	3	South
2	2023-03-01 00:00:00	Product A	220	5	North
3	2023-04-01 00:00:00	Product C	300	6	East
4	2023-05-01 00:00:00	Product B	180	4	West
5	2023-06-01 00:00:00	Product A	210	5	North
6	2023-07-01 00:00:00	Product C	320	7	East
7	2023-08-01 00:00:00	Product B	160	3	South
8	2023-09-01 00:00:00	Product A	230	6	North
9	2023-10-01 00:00:00	Product C	310	7	East
10	2023-11-01 00:00:00	Product B	190	4	West
11	2023-12-01 00:00:00	Product A	240	6	North
12	13-01-2023	Product C	330	8	East
13	14-01-2023	Product B	170	3	South
14	15-01-2023	Product A	250	7	North
15	16-01-2023	Product C	340	8	East

```
df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)
df = df.drop_duplicates().dropna()
print(df)
```

	Date	Product	Sales	Quantity	Region
0	2023-01-01	Product A	200	4	North
1	2023-02-01	Product B	150	3	South
2	2023-03-01	Product A	220	5	North
3	2023-04-01	Product C	300	6	East
4	2023-05-01	Product B	180	4	West
5	2023-06-01	Product A	210	5	North
6	2023-07-01	Product C	320	7	East
7	2023-08-01	Product B	160	3	South
8	2023-09-01	Product A	230	6	North
9	2023-10-01	Product C	310	7	East
10	2023-11-01	Product B	190	4	West
11	2023-12-01	Product A	240	6	North
12	2023-01-13	Product C	330	8	East
13	2023-01-14	Product B	170	3	South
14	2023-01-15	Product A	250	7	North
15	2023-01-16	Product C	340	8	East

```
df.groupby('Product')['Sales'].sum().plot(kind='bar')
plt.title('Sales by Product')
plt.show()
```



```
df.groupby('Region')['Sales'].sum().plot(kind='line', marker='o')
plt.show()
```

